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Agentic Autoresearch for CT Reconstruction

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ABSTRACT

Background. Deep learning has produced a large and diverse body of CT reconstruction methods. Fair comparison across them remains labor-intensive and largely manual. Moreover, many benchmarks rely on idealized data. **Purpose.** We test two things. First, can a large language model (LLM) agent do the labor of reconstruction research — implement, tune, and benchmark many methods on its own? Second, does a ranking measured on ideal data predict how methods behave under realistic noise? **Methods.** We built an agentic loop. The agent edits a solver, runs a short job on a compute cluster, reads one frozen metric, and revises. The metric is a calibrated headroom score, $hr = \max(0, 1 - \text{RMSE}/\text{RMSE}_{\text{FBP}})$, computed inside the field of view. We grounded every method in the same differentiable fan-beam projector. We benchmarked 26 reconstruction methods on two problems: Mayo low-dose CT (noise-limited) and a 128-view sparse-view breast task from the noiseless DL-Sparse-View Challenge. We selected each method's best iteration on a held-out test set and report the per-case mean $\pm$ standard deviation. To investigate the effect of noise in the noiseless breast CT challenge, we then re-evaluated every trained model on the same breast test cases with mild Poisson noise added to the inputs ($I_0 = 10^5$ photons), *without any retraining*. **Results.** Honest benchmarking gives a small tier of statistically indistinguishable top methods, not one winner. From the 26 methods the agent recombined a compact solver of ~ 195 parameters that reaches the top accuracy tier at $\sim 2\%$ of its parameters. The best compact architecture is problem-dependent: a denoiser for noisy Mayo, a filtered data-consistency and learned primal-dual unroll for sparse-view breast. Mild input noise nearly inverts the breast ranking. The noiseless champion (supervised image denoiser, $hr = 0.89$) collapses to last ($hr = 0.00$). A learned primal-dual method rises from mid-pack to champion ($hr = 0.72 \rightarrow 0.93$). Physics-regularized and hand-crafted-smoothing methods rise; supervised image-domain denoisers collapse. **Conclusions.** An LLM agent can implement, tune, and benchmark reconstruction methods at scale under a fixed metric and compute budget. It did not invent new methods, and it required some human guidance to develop highly efficient parameter methods. More importantly, a leaderboard measured on ideal data does not predict robustness — it can invert under a small, realistic perturbation. We argue that benchmarks must include noise and other perturbations by default, and that agentic autoresearch makes such multi-condition benchmarking ready for routine use.

1 | Introduction

Deep learning has changed CT reconstruction. A decade of work has produced many learned reconstructors. Examples are unrolled iterative networks, learned primal-dual schemes, variational networks¹, image-domain and dual-domain denoisers, diffusion priors (generative models, unrelated to physical diffusion),

and per-scene implicit representations.^{1,2,3,4,5} Each promises a good image from fewer views or a lower dose than filtered back-projection (FBP). The best of them deliver. At first glance the research program looks simple: pick the strongest network, train it, reconstruct.

Yet, a purely data-driven reconstructor learns a map that is not tied to the physics. Such a map is free to *hallucinate*.⁶ It can add anatomy that the measurements do not support.⁷ This is not a small tuning issue. It is a structural risk. The field's durable

¹ A variational network (Hammernik et al.) is an unrolled network whose learned regularizer is a filter-bank “field of experts”; we use it here as the accuracy yardstick.

answer is to put the physics back in. We embed the **known operator** — the differentiable forward and back projector — into the learned pipeline.^{8,9} In fact, this trade is quantified. Constraining a network with an exact operator provably cannot raise, and usually lowers, the maximum error bound. In practice it also lowers the number of parameters and the amount of training data needed.⁸ A recent deep risk estimator establishes this benefit on theoretical grounds.¹⁰

There is a second problem, and it sits one level up. The body of methods is itself the product of slow, largely manual work. The reconstruction problem can be framed as a learned inverse problem,⁵ but to compare or improve any method, a researcher must implement a solver, wire it to the data and geometry, launch a job, read a metric, form a hypothesis, change one thing, and repeat. This takes days per method. Recent “autoresearch” ideas imagine automating this loop — an automated research cycle in which the software itself proposes a change, runs the experiment, reads the result, and iterates.¹¹ Now, large language model (LLM) agents offer game-changing capacities. A large language model is a neural network trained to predict text. An *agent* wraps such a model in a loop so that it can act — read files, write code, and launch jobs — not just answer questions; a loop driven this way we call *agentic*. General-purpose coding agents already resolve real software-engineering tasks,^{12,13,14} and physics-informed agentic development has begun in adjacent imaging fields.¹⁵ *Can an LLM agent do reconstruction research?* Not just tune one number, but implement an unfamiliar method, diagnose why a reconstruction fails, edit its own solver code, and benchmark the result fairly — if we ground it in the same two things that would also guide a human: a differentiable projector and a fair, frozen metric.

We built such a loop and let an agent run it. With it, we benchmarked 26 methods across two CT problems. One is noise-limited (Mayo low-dose¹⁶); the other is incompleteness-limited (128-view sparse-view breast, from the noiseless DL-Sparse-View Challenge¹). We then asked a second question that the ideal-data benchmark cannot answer on its own: does the ranking survive a little noise? To answer it, we added mild Poisson noise to the test inputs and re-scored every trained model without retraining.

This paper makes five contributions. (i) An agentic CT-reconstruction loop grounded in a differentiable projector and a frozen calibrated metric, with full provenance. (ii) An autonomous implementation and fair benchmark of 26 methods across two regimes. (iii) An evidence-derived compact solver, competitive with the top tier at ~1–2 % of its parameters, whose optimal form is problem-dependent. (iv) A robustness result — ideal-data rank does not predict, and can invert under, noise — and the practical lesson it carries for how we build benchmarks. (v) We release all 26 methods and the agentic loop as open source (<https://github.com/akmaier/Agent4CT>).

2 | Materials and Methods

2.1 | The agentic loop

The agent works in a fixed cycle. First it reads the previous result and names the failure. It then changes one thing — a hyperparameter, or the solver code itself — and states a hypothesis. A single short cluster job runs under a fixed compute budget. The

agent reads the frozen metric and accepts or discards the change. The cycle then repeats.

Every run is grounded in the same *differentiable* fan-beam projector (PYRO-NN¹⁷) — a forward/back-projector built so that gradients flow through it, so it can sit inside a trainable network and be optimized by back-propagation. Each run also writes an immutable record: the configuration, the reconstruction, the metric, and a comparison figure. We enforced a fixed wall-clock budget per iteration (20 min on breast; matched on Mayo) so that all methods are compared under equal compute. A human coordinator supervised the loop: setting targets, redirecting across problems, and auditing for padding and provenance gaps (Section 4). The loop is sketched in Figure 1.

2.2 | A single framework for 26 methods

We describe the forward model first. For the reconstruction, the scanner is assumed as a linear operator. The sinogram is the set of line integrals of the image:

$$\mathbf{g} = \mathbf{A} \mathbf{x} + \boldsymbol{\varepsilon}. \quad (1)$$

Here $\mathbf{x}$ is the image, $\mathbf{A}$ the discrete fan-beam projector, $\mathbf{g}$ the measured projections, and $\boldsymbol{\varepsilon}$ the noise. For the noiseless Sidky breast data $\boldsymbol{\varepsilon} = \mathbf{0}$. Reconstruction means recovering $\mathbf{x}$ from $\mathbf{g}$. With 128 views, $\mathbf{A}$ is under-determined.

Almost every method is one instance of the same regularized inversion. We trade fit to the measurements against a prior:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{A} \mathbf{x} - \mathbf{g}\|_2^2 + \lambda \mathcal{R}(\mathbf{x}). \quad (2)$$

A method is fixed by two choices. The first is the **data-consistency operator** $\mathbf{D}$, which maps the measurement residual back to image space and so pulls the image toward agreement with the sinogram. The second is the **prior** $\mathcal{R}$ (also called the regularizer): it encodes what a plausible image looks like and sets the fit-versus-smoothness trade-off. Many solvers approximate Eq. (2) by an *unrolled* proximal-gradient scheme of K steps. Unrolling fixes the iteration to K steps and makes each step a trainable network layer, so the whole fixed-length scheme is trained end-to-end:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \mathbf{D}(\mathbf{A} \mathbf{x}_k - \mathbf{g}) + \mathcal{R}_{\theta}(\mathbf{x}_k). \quad (3)$$

Each step is a proximal-gradient update: a gradient step toward data fit through $\mathbf{D}$, then a “proximal” step — the learned prior $\mathcal{R}_{\theta}$ — that cleans the image. The form of $\mathbf{D}$ matters. Here, the raw adjoint $\mathbf{D} = \mathbf{A}^T$ back-projects the residual and re-injects sparse-view streaks, whereas the filtered map $\mathbf{D} = \text{FBP}(\cdot)$ (a ramp/Hann-filtered back-projection) does not. This distinction is the key lever on the sparse-view problem.

Some methods add a learned dual in the measurement domain. This is the learned primal-dual (LPD) block.² It learns updates in both the image domain (the *primal*) and the measurement/sinogram domain (the *dual*). A sinogram-space memory $\mathbf{h}$ and the image $\mathbf{x}$ update together:

$$\mathbf{h}_{k+1} = \Gamma_{\phi}(\mathbf{h}_k, \mathbf{A} \mathbf{x}_k - \mathbf{g}), \quad \mathbf{x}_{k+1} = \Lambda_{\theta}(\mathbf{x}_k, \mathbf{D}(\mathbf{h}_{k+1})). \quad (4)$$

TABLE 1 | Solver taxonomy (condensed). Each family is placed on Axis A (physics engagement, i.e. how the operator **A** is used at inference) and Axis B (prior source $\mathcal{R}$), with a source citation per family. The full per-solver configuration and parameter counts are in Supplement S1.

Family	Axis A	Axis B	Ref.
Classical iterative (TV, PWLS-TV)	in-loop	hand-crafted	19,20
Image-domain denoiser (bilateral, U-Net)	none	hand-cr. / sup.	21,22
Unrolled learned-iterative (ITNet, U-Swin)	in-loop	supervised	4,23,24,25,26
Variational network (Hammernik)	in-loop	supervised	27,3
Learned primal-dual (LPD)	in-loop	supervised	2
Dual-domain (DD sup., bilateral, N2I)	single / in-loop	sup. / self-sup.	28,29
Generative prior (fastdiff $\times 4$)	in-loop	generative	30,31,32,33
Per-scene implicit (NAF, R^2 -Gaussian)	per-scene	implicit	34,35,36
Foundation zero-shot (RAM)	single	foundation	18
Band decomposition (Wu-2015)	in-loop	hand-cr. / sup.	20
Recombination (ours)	in-loop	hybrid	—

The convolutions $\Gamma_\phi, \Lambda_\theta$ are small and weight-tied. We zero-initialize their final layer, so the block is the identity at the start and cannot regress.

Table 1 places all 26 methods on two axes. Axis A is physics engagement: how much **A** is used at inference (none, a single data-consistency step, an in-loop unrolled scheme, or a full per-scene fit). Axis B is the prior source, and its categories are worth naming plainly. *Hand-crafted* priors (total variation, bilateral) encode smoothness by hand. *Supervised* priors are trained on paired examples with a known answer (low-dose input, full-dose truth). *Self-supervised* priors train without a clean reference, using structure in the data itself; Noise2Inverse (N2I), for example, splits the measured projections into subsets so one subset’s reconstruction predicts another’s. *Generative* priors are diffusion models — networks that make realistic images by reversing a gradual noising process — used to “imagine” plausible detail. *Implicit/per-scene* methods fit a small network to each single scan that returns the attenuation at any queried coordinate, rather than storing a pixel grid; neural fields and Gaussian splatting (which models the volume as many small 3-D Gaussian blobs) are examples. A *foundation model* is a large network pre-trained once on generic data and applied here with no task-specific training (“zero-shot”); we include the Reconstruct Anything Model.¹⁸ This table also names the agent’s job in the compact study: re-select the pair $(\mathbf{D}, \mathcal{R})$. Full per-solver configurations are in Supplement S1.

Search protocol. Each of the 25 established methods was tuned by 20 autoresearch iterations of the loop in Section 2.1. The 26th method — our compact solver — was searched for 40 iterations, with the explicit aim of an ultra-low-parameter recombination of the pair $(\mathbf{D}, \mathcal{R})$. During its first 20 iterations the agent worked alone and was free to recombine any component of the other 25 solvers. During the second 20, a human operator added mild suggestions about directions not yet tried. This was the only stage where the human shaped the search beyond setting targets and auditing.

2.3 | Datasets and geometry

Mayo low-dose CT. Real helical low-dose data from the 2016 AAPM Low-Dose CT Grand Challenge.¹⁶ Rebinning helical to fan-beam via single-slice rebinning³⁷ allows direct use of the fan-beam projector. We built and validated this pipeline, and use a fixed train/val/test split (Supplement S2). This is one of a family of AAPM reconstruction grand challenges spanning spectral, truth-based, and metal-artifact tasks.^{38,39,40}

Breast CT Challenge. 128-view 2-D fan-beam sparse-view data of synthetic breast phantoms.¹ This data is **noiseless by design**. The challenge asks whether deep learning can solve the sparse-view inverse problem, so its RMSE floor is exactly zero. We re-partitioned the public 4000-case train pool into train (3600), validation (200), and test (200), with a fixed seed and disjoint cases. To confirm the split, we verified disjointness by image hashing: no test image appears in train or validation (Supplement S3).

2.4 | The calibrated-headroom metric

We score how far a method closes the gap between the low-dose FBP and the truth. Let $\mathbf{M}$ be a field-of-view mask. The headroom is

$$\text{hr} = \max\left(0, 1 - \frac{\text{RMSE}(\mathbf{M} \odot \hat{\mathbf{x}}, \mathbf{M} \odot \mathbf{x})}{\text{RMSE}(\mathbf{M} \odot \mathbf{x}_{\text{FBP}}, \mathbf{M} \odot \mathbf{x})}\right). \quad (5)$$

An hr of 0 means no better than FBP; an hr near 1 means near-perfect. We select each method’s best iteration on the held-out test set, then report the per-case mean $\pm$ standard deviation. SSIM and PSNR use a batch-wide data range so that they are comparable across cases.

2.5 | The noise-robustness experiment

We ask whether the noiseless ranking survives a little input noise. For each method we take its noiseless best iteration. We then re-evaluate it on the **same** 200 breast test cases, but with photon noise added to the sinograms. For a line integral p and incident photon count I_0 ,

$$N \sim \text{Poisson}(I_0 e^{-p}), \quad \hat{p} = -\log(\max(N, 1)/I_0). \quad (6)$$

We use $I_0 = 10^5$ photons. This is a rather high dose. The noise is mild — about 1–2 % at the thickest ray. We apply it to the test inputs only. Note that the ground truth stays clean. Supervised solvers **load their noiseless-trained weights and no additional training is applied**. Per-scene and classical solvers re-fit on the noisy input, which is their normal inference. The FBP baseline in Eq. (5) is recomputed from the same noisy data, so hr measures improvement over the noisy FBP. This is exactly the realistic model $\mathbf{Ax} + \varepsilon$ that the noiseless challenge does not study.

2.6 | Statistics

We begin with Mayo. There the five held-out patients give $n = 5$, so the paired t -test has low power; we therefore report both the 5 % and 1 % levels. For Breast CT, the 200 test cases are shared by all methods, so comparisons are paired. With $n = 200$ the paired t -test is very high-powered: tiny mean gaps reach $p < 0.01$. Hence, we report **effect size** — how large a difference is, independent of sample size — not just p . We use Cohen’s $d_z = \bar{d}/s_d$, the mean of the paired differences over their standard deviation, and lead with d_z and the raw difference. The contrast between $n = 5$ and $n = 200$ is discussed in Section 4.

3 | Results

3.1 | Mayo low-dose CT: a tier, not a winner

On Mayo, the test-selected leaderboard does not yield a single winner. It yields a small top tier that is statistically indistinguishable. The champion is ITNet at hr 0.376 ± 0.089 , with ITNet-v2 (0.374) and U-Swin (0.370) inside one standard error. A paired t -test cannot separate ITNet, its variants, and U-Swin at the 5% level, and our compact recombination solver (Section 3.3) joins this tier at the 1% level ($p = 0.02$ – 0.027). The reason is sample size: the Mayo test set has five patients, so $n = 5$, and at that size the paired t -test has little power while a Wilcoxon signed-rank test cannot reach $p < 0.05$ for any pair. Hence the honest reading of the top of the board is a tie, not a strict order. Note that the absolute image metrics agree with this reading. The top six solvers all exceed SSIM 0.97 and sit at the RMSE floor of 5×10^{-4} , so SSIM saturates exactly where the calibrated headroom still separates methods. Below the tier the board spreads cleanly: the variational network reaches hr 0.159, classical total variation 0.096, and the PWLS-TV and bilateral baselines fall further. Several families — diffusion priors, per-scene implicit methods, the zero-shot foundation model, and the primal-dual unroll as configured for the Mayo geometry — do not clear the low-dose FBP baseline and sit at hr 0, even though their SSIM stays high (0.82–0.95); SSIM rewards the smoothness that hr, measured against FBP, does not credit. The full board, the per-patient headroom, and the pairwise significance matrix are in Table 2 and Supplement S4. One practical result deserves emphasis, and it concerns effort. We can read it from the version-control history. The dominant *human* cost was the one-time helical-to-fan geometry pipeline, which the agent used but did not build; the benchmarking itself needed almost no hands-on work. The three-week benchmark campaign produced 443 commits, of which 439 are automated per-iteration provenance the agent wrote — only four were human-authored. Across the whole project the split is similar: of $\sim 1,780$ commits, roughly 1,270 are machine-written. The agent, meanwhile, ran about 1,100 short reconstruction iterations totalling ~ 134 GPU-hours of cluster compute (mean ~ 4 minutes each), executed largely in parallel. Human involvement was bursty — launching or nursing a control loop, reading a board, giving a hint — so the multi-week calendar span, stretched further by the supervisor's travel, reflects the one-time pipeline and not continuous effort. The wall-clock work thus splits sharply: many machine-hours of autonomous agent compute against scattered, intermittent human oversight.

3.2 | Noiseless Breast CT Challenge: the numbers are not a leak

On the noiseless Breast CT Challenge the headroom values are high. The champion is the supervised dual-domain denoiser at hr 0.89, with the ITNet family (0.87–0.89) and U-Swin (0.86) just behind; the top five solvers all exceed 0.85 (Table 3, left board). This is a property of the data, not a flaw and not a leak. The Breast CT Challenge data is noiseless by design, so its RMSE floor is exactly zero and near-perfect recovery is possible in principle. Because such numbers invite suspicion, we ruled out train/test leakage three independent ways. First, by mechanism: the training loader only ever reads the train split, and the test redirect fires solely at evaluation, so no model sees a test image while training. Second,

by content: over all 3600 train, 200 validation, and 200 test truth images, a pixel-hash comparison finds zero overlap in every direction. Third, by behavior: for every method the test headroom tracks the validation headroom to within about ± 0.01 , and several methods score *lower* on test than on validation. Training on the test set would push test-hr well above val-hr; it does not. Hence the high numbers reflect an easy, well-posed, incompleteness-limited problem, not memorization. This also means the breast and Mayo hr scales are not comparable. Breast is limited by missing views (128 of ~ 1000), Mayo by real quantum noise, so a breast hr of 0.89 and a Mayo hr of 0.38 describe different bottlenecks, not different amounts of the same thing (Supplement S3).

3.3 | A compact, problem-dependent solver

Beyond ranking the 25 established methods, the agent recombined their parts into a compact solver — and the best recombination is problem-dependent. On Mayo, a noise-limited problem, the compact optimum is a small multi-scale bilateral denoiser of about 970 parameters (hr 0.324) that joins the top tier at the 1% level; where the bottleneck is noise, a tiny denoiser suffices. On breast, a sparse-view problem, the optimum is a different object entirely. It is a filtered data-consistency step ($\mathbf{D} = \text{FBP}$ applied to the residual $\mathbf{A}\hat{\mathbf{x}} - \mathbf{g}$), a learned cross-step combination in the primal-dual style (Eq. 4), and a small multi-scale bilateral filter on the output. The climb to it was made of single-knob iterations, each earned against the frozen metric. A pure image-domain denoiser, however configured, capped at hr ≈ 0.26 . Replacing the adjoint by the *filtered* gradient broke that ceiling with only 13 parameters (hr 0.43). A learned cross-step combination added headroom (183 parameters, hr 0.50). A three-scale bilateral output filter completed the design (195 parameters). The final breast solver reaches hr 0.6212 ± 0.0076 — the variational-network accuracy tier at $\sim 2\%$ of that network's parameters, and with the tightest per-case standard deviation of any solver on the board. The two compact solutions do not transfer. On a sparse-view problem data-consistency is the differentiator; on a noise-limited problem it is not. This problem-dependence is the practical face of the known-operator principle: embedding the exact forward operator cannot raise, and usually lowers, the maximum error bound, and it lets a few hundred parameters do the work of millions.⁸ In fact, the agent re-derived each optimum from its own runs; when steered toward the Mayo answer on breast it did not transfer, because the evidence on breast pointed elsewhere (Supplement S5). Figure 2 places both compact solvers against parameter count.

3.4 | A little noise inverts the ranking

We now perturb the inputs and re-score, with no retraining. We add mild Poisson noise to the 200 breast test sinograms at $I_0 = 10^5$ photons — about 1–2% at the thickest ray — keep the ground truth clean, reload each model's noiseless-trained weights, and recompute the FBP baseline from the noisy data so that hr measures improvement over the *noisy* FBP. Nothing else changes. The ranking nearly inverts. Of the five best solvers on clean data, only ITNet-v3 survives in the noisy top five; the rest are displaced by physics-regularized and hand-crafted-smoothing methods. Table 3 shows the two boards side by side. The headline entries:

TABLE 2 | Mayo low-dose CT test board (all 26 solvers; $n = 5$ test patients). hr is the calibrated headroom (Eq. 5) and the ranking metric; SSIM and RMSE use a batch-wide data range (RMSE in line-integral units). Params are trainable-parameter counts; entries suffixed M are in millions, an unsuffixed number is a raw count. “–” marks a solver below the ranking floor (hr ≤ 0). Solver keys abbreviated: dd = dual-domain, px = pixel, wav = wavelet, dc = data-consistency step, con./unc. = (un)constrained, sup = supervised.

Rank	Solver	hr	SSIM	RMSE	Params
1	itnet	0.3756	0.9790	0.0005	0.233 M
2	itnet-v2	0.3735	0.9784	0.0005	0.233 M
3	uswin	0.3700	0.9770	0.0005	3.954 M
4	dd-supervised	0.3607	0.9758	0.0005	0.466 M
5	param-efficient	0.3241	0.9727	0.0005	969
6	itnet-v3	0.3066	0.9707	0.0006	3.699 M
7	hammernik-vn	0.1591	0.9520	0.0007	0.005 M
8	tv-iterative	0.0959	0.9642	0.0007	0
9	hammernik-2017	0.0843	0.9358	0.0007	0.012 M
10	manhart-pwls-tv	0.0624	0.9649	0.0007	0
11	wu-2015-trainable	0.0207	0.8923	0.0008	8
12	dd-n2i	0.0176	0.9590	0.0008	0.466 M
13	manduca-bilateral	0.0159	0.9561	0.0009	7

Rank	Solver	hr	SSIM	RMSE	Params
14	dd-bilateral-sup	0.0040	0.9541	0.0009	12
–	tv-iterative-sup	0.0109	0.9366	0.0008	2
–	dd-bilateral-n2i	0.0000	0.9587	0.0010	12
–	fastdiff-flow-px-unc	0.0000	0.9523	0.0012	3.823 M
–	fastdiff-wdm-wav-con	0.0000	0.9501	0.0013	3.825 M
–	ram	0.0000	0.9446	0.0012	35.619 M
–	learned-primal-dual	0.0000	0.9415	0.0011	0.155 M
–	fastdiff-wdm-wav-unc	0.0000	0.9404	0.0013	3.825 M
–	fastdiff-flow-px-con	0.0000	0.9300	0.0013	3.823 M
–	diff-recon-dc-unc	0.0000	0.9084	0.0014	8.594 M
–	diff-recon-dc-con	0.0000	0.8766	0.0016	8.594 M
–	r2gaussian	0.0000	0.8234	0.0043	0.003 M
–	naf	0.0000	0.8172	0.0028	33.558 M

- **learned-primal-dual**² rises from hr 0.72 (rank 6) to **0.93** (rank 1), with noisy SSIM 0.99. It keeps the forward operator in the loop, so it re-solves the noisy measurements rather than denoising a fixed image.
- **manduca-bilateral**, a 22-parameter hand-crafted smoothing prior,²¹ rises from 0.28 to **0.84** (rank 2). It has nothing to overfit, so smoothing is noise-robust by construction. The self-supervised dual-domain bilateral (dd-bilateral-n2i)²⁹ rises even more sharply, from an essentially failed 0.0001 on clean data to **0.79** (rank 3).
- **dual-domain-supervised**, the noiseless champion (0.89, rank 1),²⁸ **collapses to hr 0.00** and out of the ranking, with SSIM 0.35. It is a pure supervised image-domain denoiser trained only on clean FBP, so a distribution it never saw breaks it.
- The supervised unrolled family (ITNet v1/v2/v3, U-Swin) drops from 0.86–0.89 into the middle of the board (0.55–0.70). Our compact solver holds relatively well (0.62 → 0.52, rank 13).
- Most strikingly, the unconstrained wavelet diffusion prior (fastdiff-wdm-wav-unc) — a deliberate negative control that *failed* on clean data (hr 0, below the FBP floor) — is noise-robust and ranks 8th (hr 0.64, SSIM 0.95). Its generative prior never fit the clean forward model, so it degrades gracefully; the zero-shot foundation model (RAM)¹⁸ rises for the same reason (rank 7).
- Per-scene neural-field and Gaussian-splatting methods (NAF, R²-Gaussian)^{34,35} do not complete under the 20-minute budget on either board and are reported as DNF.

The absolute SSIM and PSNR columns confirm the reordering is genuine, not an artifact of the changing FBP baseline: learned-primal-dual reaches SSIM 0.99 on the noisy data while dual-domain-supervised falls to 0.35 (Supplement S6). Because every method sees the identical noisy input and shares the same noisy FBP baseline, the comparison is fair within the noisy board. Figure 3 shows the full reordering as a paired rank plot, and Figure 4 shows its pixel-level face for the two methods at opposite ends.

3.5 | The framework explains the reversal

The reversal is not a surprise once we read it off the two axes of Table 1. Brittleness concentrates in one corner: supervised priors with weak physics engagement — image-domain maps trained on clean FBP. Dual-domain-supervised is the extreme case. It applies a learned image-to-image map with no forward operator at inference, so a small input shift it never saw during training moves it off its learned manifold and the reconstruction collapses. The constrained diffusion variants, which also lean on a clean-data fit, fall the same way. Robustness concentrates where the prior is not tuned to the clean distribution, and two disjoint mechanisms reach it. The first is in-loop physics: learned-primal-dual and the variational networks keep the forward operator **A** in the loop, so they re-solve the noisy measurements instead of denoising a fixed image, and their ranking rises or holds. The second is a prior that never fit the clean data at all: hand-crafted smoothing (bilateral, total variation) has nothing to overfit, and even the generative and foundation priors that failed on clean data degrade gracefully precisely because they never matched the clean forward model. As such, the framework turns the reversal from a curiosity into a prediction. A method’s position on the physics-engagement and prior-source axes tells us, before any noise is added, whether it is likely to be brittle or robust. The lesson for benchmarking follows directly: a leaderboard measured on a single, idealized distribution rewards exactly the methods most likely to fail under a small, realistic perturbation.

4 | Discussion

4.1 | What the agent did well, and what it did not

We give an honest scorecard. It is the core of this Discussion.

The agent is strong in four ways. First, it is fast from paper to code: it turns a method description into a working solver quickly. This is the main speed-up, and it is why 26 methods across two datasets was feasible. In fact, it is also strong at hyper-parameter optimization. It respects a fixed compute budget, which is hard for humans and easy for the agent. And it scales evaluation: running many benchmarks in parallel is its strongest practical use.

TABLE 3 | The two breast boards side by side, for the same trained models on the same 200 test cases with *no retraining*. Left: noiseless board. Right: noisy board ($I_0 = 10^5$). The ranking nearly inverts. hr is the calibrated headroom (Eq. 5) and the ranking metric; SSIM and RMSE use a batch-wide data range (RMSE in line-integral units). “—” marks a solver that did not clear the ranking floor ($hr \leq 0$) or did not finish (—). Params (M suffix = millions, otherwise a raw count) and solver keys as in Table 2.

Rank	Solver	hr	SSIM	RMSE	Params
1	dd-supervised	0.8948	0.9992	0.0005	0.466 M
2	itnet	0.8926	0.9991	0.0006	0.233 M
3	itnet-v2	0.8893	0.9991	0.0006	0.928 M
4	itnet-v3	0.8749	0.9989	0.0007	8.318 M
5	uswin	0.8586	0.9986	0.0007	7.025 M
6	learned-primal-dual	0.7233	0.9962	0.0014	0.304 M
7	hammernik-2017	0.6265	0.9902	0.0019	0.009 M
8	param-efficient	0.6212	0.9912	0.0020	195
9	hammernik-vn	0.5787	0.9865	0.0022	0.008 M
10	fastdiff-flow-px-con	0.5119	0.9792	0.0025	3.823 M
11	ram-zeroshot	0.3693	0.9787	0.0033	35.619 M
12	tv-iterative	0.3601	0.9918	0.0033	0
13	manhart-pwls-tv	0.3601	0.9918	0.0033	0
14	wu-2015-trainable	0.3156	0.9537	0.0035	12
15	manduca-bilateral	0.2763	0.9830	0.0037	22
16	fastdiff-flow-px-unc	0.2693	0.9776	0.0038	3.823 M
17	fastdiff-wdm-wav-con	0.2693	0.9340	0.0038	3.825 M
18	dd-bilateral-sup	0.2546	0.9848	0.0039	24
19	wu-2015	0.1517	0.9611	0.0044	0
20	tv-iterative-sup	0.1513	0.9613	0.0044	20
21	dd-bilateral-n2i	0.0001	0.9583	0.0061	6
—	dd-n2i	0.0001	0.9461	0.0062	0.466 M
—	fastdiff-wdm-wav-unc	0.0000	0.9448	0.0112	3.825 M
—	naf	—	—	—	33.563 M
—	r2gaussian	—	—	—	0.006 M

Rank	Solver	hr	SSIM	RMSE	Params
1	learned-primal-dual	0.9349	0.9905	0.0020	0.304 M
2	manduca-bilateral	0.8420	0.9537	0.0049	22
3	dd-bilateral-n2i	0.7927	0.9532	0.0064	6
4	hammernik-2017	0.7016	0.7178	0.0092	0.009 M
5	itnet-v3	0.7007	0.7564	0.0092	8.318 M
6	fastdiff-flow-px-unc	0.6620	0.7058	0.0104	3.823 M
7	ram-zeroshot	0.6382	0.6578	0.0111	35.619 M
8	fastdiff-wdm-wav-unc	0.6350	0.9450	0.0112	3.825 M
9	uswin	0.5915	0.6408	0.0125	7.025 M
10	itnet	0.5784	0.6295	0.0129	0.233 M
11	hammernik-vn	0.5529	0.5962	0.0137	0.008 M
12	itnet-v2	0.5480	0.6201	0.0139	0.928 M
13	param-efficient	0.5153	0.5771	0.0149	195
14	manhart-pwls-tv	0.5068	0.6164	0.0151	0
15	tv-iterative	0.5068	0.6164	0.0151	0
16	wu-2015	0.3700	0.5244	0.0193	0
17	wu-2015-trainable	0.2591	0.4400	0.0227	12
18	tv-iterative-sup	0.1071	0.4567	0.0274	20
19	dd-bilateral-sup	0.1003	0.4524	0.0276	24
—	dd-supervised	0.0000	0.3483	0.0338	0.466 M
—	fastdiff-wdm-wav-con	0.0000	0.3104	0.0477	3.825 M
—	fastdiff-flow-px-con	0.0000	0.2769	0.0583	3.823 M
—	dd-n2i	—	—	—	0.117 M
—	naf	—	—	—	33.563 M
—	r2gaussian	—	—	—	0.006 M

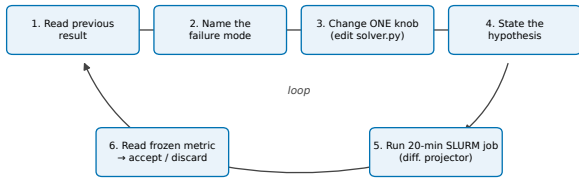


FIGURE 1 | The agentic autoresearch loop and its provenance. The agent reads the previous result, names the failure, changes one thing, states a hypothesis, runs one short cluster job grounded in the same differentiable fan-beam projector, reads the frozen calibrated-headroom metric, and accepts or discards the change. Every iteration writes an immutable record (configuration, reconstruction, metric, comparison figure).

The agent is weak in several ways. It does not invent new methods. Rather, it implements, tunes, and recombines known ones, and proposed no new reconstruction principle. Yet, it had to be *forced* to mix and match. The idea of combining proven pieces into one compact solver was a human idea, on both datasets. The agent executed it well but did not originate it. Left alone, it converges early and stops exploring. The compact-solver search shows this directly: the agent’s own first 20 iterations plateaued at $hr \approx 0.50$, and the decisive gains — the break to $hr 0.62$ — came only after a human began suggesting untried directions from iteration 20 on (Supplement S5). Its CT-image vision is unreliable: it keeps missing very clear artifacts in slices and sinograms, so numbers had to be the source of truth. Long tasks must be

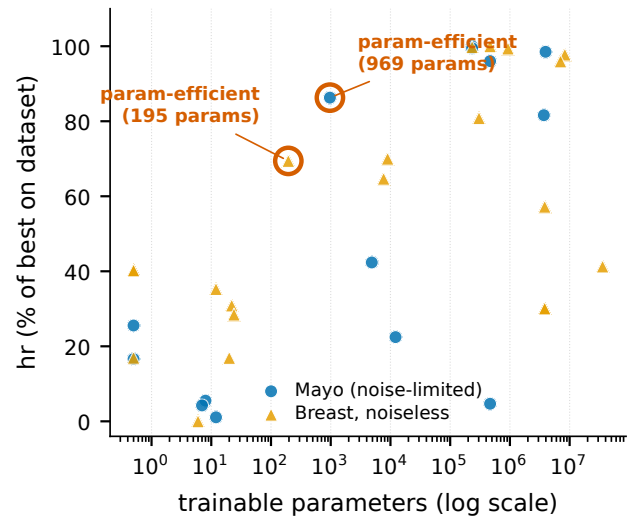


FIGURE 2 | Accuracy versus parameter count for both datasets (Mayo and breast noiseless). To overlay the two different hr scales, each solver’s hr is normalized to the percent of the best hr reached on its own board. The agent’s compact recombination solver is ringed: 969 parameters on Mayo, 195 on breast. It reaches the top accuracy tier at $\sim 1\text{--}2\%$ of the parameters of the variational network.

decomposed into sub-tasks; otherwise even a one-million-token context is used up quickly. And it overfits to the task and metric. Of course, this last point is probably not only an agent problem — human researchers overfit to benchmarks too.

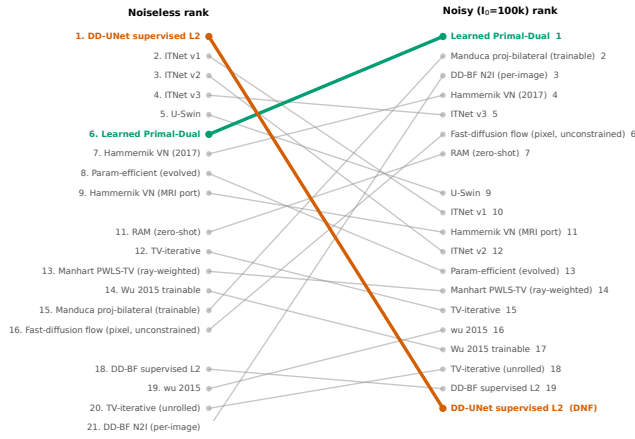


FIGURE 3 | The reversal. Paired rank plot from the noiseless breast board to the noisy breast board ($I_0 = 10^5$, same trained models, same 200 test cases, no retraining), with an arrow per method. The supervised image-domain champion collapses from rank 1 to last; the learned primal-dual method rises from mid-pack to champion; hand-crafted smoothing and in-loop physics priors rise. A small, realistic noise perturbation reorders almost the whole leaderboard.

The division of labor is clear. On one side, the agent is a tireless, budget-respecting, self-modifying executor. On the other, the human remains the strategist and auditor — forcing breadth, supplying the recombination idea, redirecting across problems, decomposing the work, and checking for padding and provenance. That division is the honest answer to the title question.

4.2 | Two scientific findings

First, the best compact architecture is problem-dependent. The Mayo solution is a denoiser, whereas the breast solution is a filtered-data-consistency and primal-dual unroll. Again, the agent re-derived each; it did not transfer one to the other. This supports the known-operator view: the right inductive bias — the assumptions an architecture builds in before it sees any data — depends on whether the problem is noise-limited or incompleteness-limited.

Second, a little noise broke the whole breast leaderboard. The methods that won on clean data were the least robust. This is a caution for the field. Significance also depends on sample size: at $n = 5$ (Mayo) the top methods tie; at $n = 200$ (breast) everything separates. As such, p -values are not comparable across datasets, and effect size should lead.

4.3 | Limitations

We used a single agent and a single metric. The geometry/data-engineering bottleneck remains; the agent did not remove it. Some solutions are seed-fragile. Two per-scene methods did not complete under the budget. The noise study uses one dose level; a dose sweep is future work.

5 | Conclusions

An LLM agent can implement, tune, and benchmark 26 CT reconstruction methods under one frozen metric and a fixed compute budget. The honest outcome of such a benchmark is a small tier of indistinguishable top methods, not one winner. Still, the agent does the labor; it does not replace the human strategist.

The more important message is about benchmarks. We used the agent to scale evaluation, and that scale exposed a problem. Our ideal-data leaderboard rewarded brittle methods. A small, realistic noise perturbation reordered almost the whole board. The methods that won on clean data collapsed. Hence, we argue that benchmarks must change. They should include noise, dose variation, and other realistic perturbations by default, so that leaderboards reward robust methods rather than methods that overfit ideal data. Agentic autoresearch makes such multi-condition benchmarking cheap enough to be routine. That is its real payoff for the field.

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To be completed: funding, compute, data providers — Mayo, Sidky group / AAPM DL-Sparse-View Challenge.

Conflicts of Interest

The authors declare no relevant conflicts of interest.

Data and Code Availability

Boards, provenance records, and code are at <https://github.com/akmaier/Agent4CT>; live dashboards at <https://akmaier.github.io/Agent4CT>.

References

- Sidky EY, Pan X. Report on the AAPM Deep-Learning Sparse-View CT (DL-Sparse-View CT) Grand Challenge. *Medical Physics* 2022; 49(8): 4935–4943. doi: 10.1002/mp.15489
- Adler J, Öktem O. Learned Primal-Dual Reconstruction. *IEEE Transactions on Medical Imaging* 2018; 37(6): 1322–1332. doi: 10.1109/TMI.2018.2799231
- Hammernik K, Klatzer T, Kobler E, et al. Learning a Variational Network for Reconstruction of Accelerated MRI Data. *Magnetic Resonance in Medicine* 2018; 79(6): 3055–3071. doi: 10.1002/mrm.26977
- Würfl T, Hoffmann M, Christlein V, et al. Deep Learning Computed Tomography: Learning Projection-Domain Weights From Image Domain in Limited Angle Problems. *IEEE Transactions on Medical Imaging* 2018; 37(6): 1454–1463. doi: 10.1109/TMI.2018.2833499
- Ongie G, Jalal A, Metzler CA, Baraniuk RG, Dimakis AG, Willett R. Deep Learning Techniques for Inverse Problems in Imaging. *IEEE Journal on Selected Areas in Information Theory* 2020; 1(1): 39–56. doi: 10.1109/JSAIT.2020.2991563
- Huang Y, Würfl T, Breininger K, Liu L, Lauritsch G, Maier A. Some investigations on robustness of deep learning in limited angle tomography. In: Springer. ; 2018: 145–153.
- Maier A, Syben C, Lasser T, Riess C. A Gentle Introduction to Deep Learning in Medical Image Processing. *Zeitschrift für Medizinische Physik* 2019; 29(2): 86–101. doi: 10.1016/j.zemedi.2018.12.003
- Maier AK, Syben C, Stimpel B, et al. Learning with Known Operators Reduces Maximum Error Bounds. *Nature Machine Intelligence* 2019; 1(8): 373–380. doi: 10.1038/s42256-019-0077-5

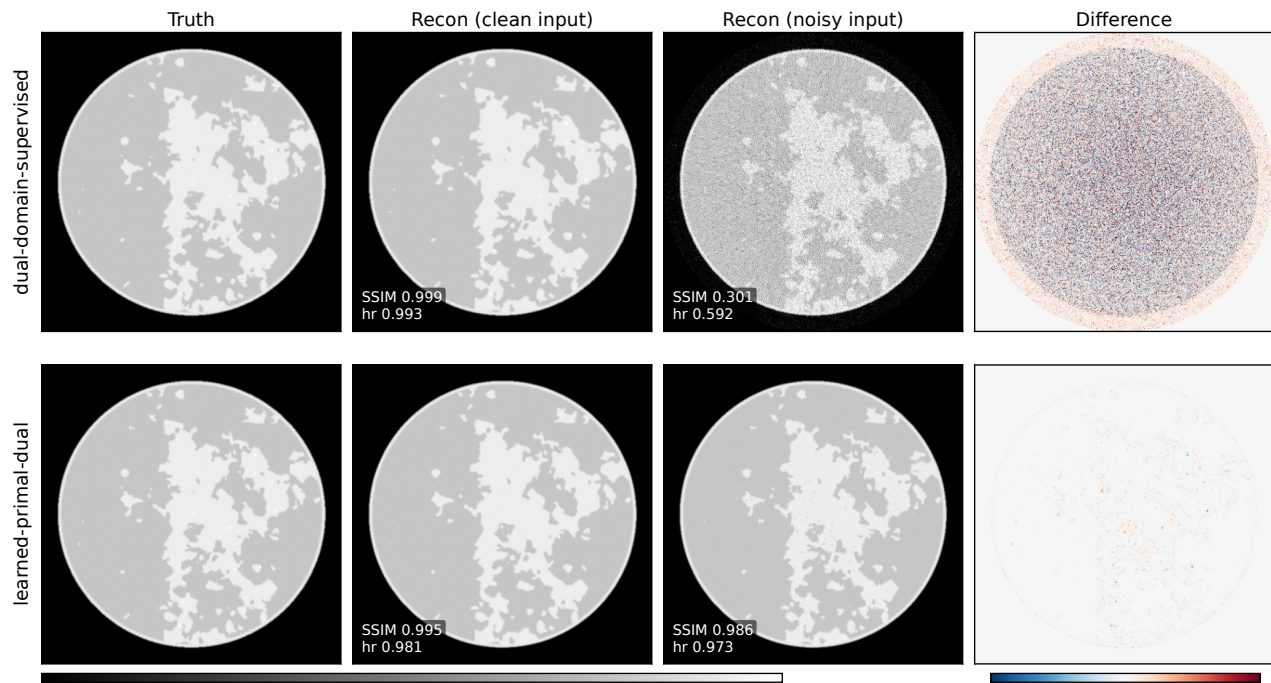


FIGURE 4 | Example reconstructions behind the reversal, for the two methods at opposite ends of it. Rows: the noiseless champion *dual-domain-supervised* (a supervised image-domain denoiser) versus *learned-primal-dual* (in-loop physics). Columns: ground truth, reconstruction from the clean input, reconstruction from the noisy input ($I_0 = 10^5$, same trained model, no retraining), and the difference to truth; SSIM is annotated on each recon. On clean inputs both are near-perfect. Under mild input noise the supervised denoiser collapses (SSIM 0.999 $\rightarrow$ 0.301), spraying structured error across the field of view, while the physics-in-the-loop primal-dual method holds (SSIM 0.995 $\rightarrow$ 0.986). This is the pixel-level face of the ranking inversion in Figure 3 and Table 3.

9. Maier A, Köstler H, Heisig M, Krauss P, Yang SH. Known Operator Learning and Hybrid Machine Learning in Medical Imaging—A Review of the Past, the Present, and the Future. *Progress in Biomedical Engineering* 2022; 4(2): 022002. doi: [10.1088/2516-1091/ac5b13](https://doi.org/10.1088/2516-1091/ac5b13)
10. Maier A, Hasan M, Conrad P, Perez-Toro PA. A Deep Risk Estimator for Known Operator Learning. *arXiv preprint arXiv:2605.08517* 2026.
11. Lu C, Lu C, Lange RT, Foerster J, Clune J, Ha D. The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery. *arXiv preprint arXiv:2408.06292* 2024. arXiv:2408.06292.
12. Yang J, Jimenez CE, Wettig A, et al. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. In: ; 2024. arXiv:2405.15793.
13. Jimenez CE, Yang J, Wettig A, et al. SWE-bench: Can Language Models Resolve Real-World GitHub Issues?. In: ; 2024. arXiv:2310.06770.
14. Maier A, Bergler C, Bayer S, Zaiss M., eds. *Vibe Coding: Software Engineering in the Age of AI*. Lecture Notes in Computer ScienceSpringer . 2026. in print.
15. Zaiss M, Aly A, Endres J, Dornstetter T, Weinmüller S, Maier A. MR LLMs with MR Agentic Sequence Development: Leveraging Skills for Automatic Physics-Informed Sequence Development. *arXiv preprint arXiv:2604.13282* 2026. arXiv:2604.13282; Agent4MR.
16. McCollough CH, Bartley AC, Carter RE, et al. Low-Dose CT for the Detection and Classification of Metastatic Liver Lesions: Results of the 2016 Low Dose CT Grand Challenge. *Medical Physics* 2017; 44(10): e339–e352. doi: [10.1002/mp.12345](https://doi.org/10.1002/mp.12345)
17. Syben C, Michen M, Stimpel B, Seitz S, Ploner S, Maier AK. Technical Note: PYRO-NN: Python Reconstruction Operators in Neural Networks. *Medical Physics* 2019; 46(11): 5110–5115. doi: [10.1002/mp.13753](https://doi.org/10.1002/mp.13753)
18. Terris M, Hurault S, Song M, Tachella J. Reconstruct Anything Model: A Lightweight Foundational Model for Computational Imaging. *arXiv preprint arXiv:2503.08915* 2025. arXiv:2503.08915.
19. Sidky EY, Pan X. Image Reconstruction in Circular Cone-Beam Computed Tomography by Constrained, Total-Variation Minimization. *Physics in Medicine & Biology* 2008; 53(17): 4777–4807. doi: [10.1088/0031-9155/53/17/021](https://doi.org/10.1088/0031-9155/53/17/021)
20. Wu M, Maier A, Yang Q, Fahrig R. A Novel Filtered Backprojection-Based Algorithm for Sparse-View CT Image Reconstruction. In: ; 2015: 202–205.
21. Manduca A, Yu L, Trzasko JD, et al. Projection Space Denoising with Bilateral Filtering and CT Noise Modeling for Dose Reduction in CT. *Medical Physics* 2009; 36(11): 4911–4919. doi: [10.1118/1.3232004](https://doi.org/10.1118/1.3232004)
22. Wagner F, Thies M, Gu M, et al. Ultralow-Parameter Denoising: Trainable Bilateral Filter Layers in Computed Tomography. *Medical Physics* 2022; 49(8): 5107–5120. doi: [10.1002/mp.15718](https://doi.org/10.1002/mp.15718)
23. Xu Y, Han S, Wang D, Wang G, Maltz JS, Yu H. Hybrid U-Net and Swin-Transformer Network for Limited-Angle Cardiac Computed Tomography. *Physics in Medicine & Biology* 2024; 69(10): 105012. doi: [10.1088/1361-6560/ad3db9](https://doi.org/10.1088/1361-6560/ad3db9)
24. Liang J, Cao J, Sun G, Zhang K, Van Gool L, Timofte R. SwinIR: Image Restoration Using Swin Transformer. In: ; 2021: 1833–1844. arXiv:2108.10257
25. Liu Z, Lin Y, Cao Y, et al. Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows. In: ; 2021: 9992–10002. arXiv:2103.14030
26. Zhang Z, Yu L, Liang X, Zhao W, Xing L. TransCT: Dual-Path Transformer for Low-Dose Computed Tomography. In: Lecture Notes in Computer Science. Springer; 2021: 55–64. arXiv:2103.00634

27. Hammernik K, Würfl T, Pock T, Maier A. A Deep Learning Architecture for Limited-Angle Computed Tomography Reconstruction. In: Informatik aktuell. Springer; 2017: 92–97
28. Wagner F, Thies M, Pfaff L, et al. On the Benefit of Dual-Domain Denoising in a Self-Supervised Low-Dose CT Setting. In: IEEE; 2023: 1–5
29. Hendriksen AA, Pelt DM, Batenburg KJ. Noise2Inverse: Self-Supervised Deep Convolutional Denoising for Tomography. *IEEE Transactions on Computational Imaging* 2020; 6: 1320–1335. doi: [10.1109/TCI.2020.3019647](https://doi.org/10.1109/TCI.2020.3019647)
30. Shi J, Pelt DM, Batenburg KJ. DM4CT: Benchmarking Diffusion Models for Computed Tomography Reconstruction. In: ; 2026. arXiv:2602.18589.
31. Friedrich P, Wolleb J, Bieder F, Durrer A, Cattin PC. WDM: 3D Wavelet Diffusion Models for High-Resolution Medical Image Synthesis. In: LNCS. Springer; 2024. arXiv:2402.19043
32. Song Y, Shen L, Xing L, Ermon S. Solving Inverse Problems in Medical Imaging with Score-Based Generative Models. In: ; 2022. arXiv:2111.08005.
33. Chung H, Kim J, McCann MT, Klasky ML, Ye JC. Diffusion Posterior Sampling for General Noisy Inverse Problems. In: ; 2023. arXiv:2209.14687.
34. Zha R, Zhang Y, Li H. NAF: Neural Attenuation Fields for Sparse-View CBCT Reconstruction. In: Lecture Notes in Computer Science. Springer; 2022: 442–452. arXiv:2209.14540
35. Zha R, Lin TJ, Cai Y, Cao J, Zhang Y, Li H. R²-Gaussian: Rectifying Radiative Gaussian Splatting for Tomographic Reconstruction. In: ; 2024. arXiv:2405.20693.
36. Liu H, Liu B, Hu Q, et al. A Review on 3D Gaussian Splatting for Sparse View Reconstruction. *Artificial Intelligence Review* 2025; 58: 215. doi: [10.1007/s10462-025-11171-4](https://doi.org/10.1007/s10462-025-11171-4)
37. Noo F, Defrise M, Clackdoyle R. Single-Slice Rebinning Method for Helical Cone-Beam CT. *Physics in Medicine & Biology* 1999; 44(2): 561–570. doi: [10.1088/0031-9155/44/2/019](https://doi.org/10.1088/0031-9155/44/2/019)
38. Sidky EY, Pan X. Report on the AAPM Deep-Learning Spectral CT Grand Challenge. *Medical Physics* 2024. arXiv:2212.06718doi: [10.1002/mp.16612](https://doi.org/10.1002/mp.16612)
39. Abadi E, Segars WP, Felice N, Hoffman EA, Samei E. The AAPM Truth-Based CT (TrueCT) Reconstruction Grand Challenge. *Medical Physics* 2025; 52(4): 1978–1990. doi: [10.1002/mp.17619](https://doi.org/10.1002/mp.17619)
40. Haneda E, Peters N, Zhang J, et al. AAPM CT Metal Artifact Reduction Grand Challenge. *Medical Physics* 2025; 52(10): e70050. doi: [10.1002/mp.70050](https://doi.org/10.1002/mp.70050)